CN Practical 2: Create and Test a Standard Straight Network Cable

Aim: To create and test a standard straight-through network cable using a crimping tool and cable tester.

Theory:

An Ethernet cable is used to establish wired communication between computers and networking devices. A straight-through cable is a type of Ethernet cable in which both ends follow the same wiring standard. The T568A and T568B standards define the colour sequence of the eight wires inside an RJ-45 connector.

Straight-through cables are commonly used to connect a computer to a switch, router or access point. UTP (Unshielded Twisted Pair) cable is widely used because it is economical, flexible and easy to install. A crimping tool is used to attach RJ-45 connectors to the cable ends, while a LAN cable tester checks whether the wires are connected correctly.

Materials required:

- UTP Ethernet cable
- Two RJ-45 connectors
- Crimping tool
- LAN cable tester
- Cable cutter or stripper

T568B colour sequence:

Pin	Wire colour
1	White-Orange
2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

Procedure:

1. Measure and cut the required length of UTP cable.
2. Remove approximately 2–3 cm of the outer insulation from both ends without damaging the internal wires.
3. Untwist and straighten the eight wires carefully.
4. Arrange the wires in the T568B colour sequence.
5. Trim the wire ends evenly so that they can enter the RJ-45 connector completely.
6. Insert the wires into the connector, ensuring that each wire reaches its proper contact.
7. Use the crimping tool to secure the connector firmly.
8. Repeat the same T568B sequence at the other end of the cable.
9. Connect both ends to a LAN cable tester.

10. Check the tester indicators to confirm that all eight connections are correct.

Observation:

The straight-through cable was prepared using the T568B standard at both ends. During testing, the cable tester should indicate continuity for pins 1 through 8 in the corresponding order. Any missing or incorrect indication may indicate a broken wire, loose connector or incorrect wiring.

Conclusion:

A standard straight-through Ethernet cable was prepared using a UTP cable and RJ-45 connectors. The cable was tested using a LAN cable tester to verify the wiring sequence and continuity. The practical demonstrated the correct procedure for preparing Ethernet cables for network connections.